

CoM3D-ACE: Reciprocal Evidence-Graph Tracklet Refinement for Aerial Multi-Object Tracking

Jong-Chan Park^a, So-Hee Lim^a, Sujin Park^a, Sang-Min Choi^{a,*},
Gun-Woo Kim^{a,*}

^a*Gyeongsang National University, Jinju, 52828, Republic of Korea*

Abstract

Aerial multi-object trackers often split one identity into several local tracklets when small targets are occluded or weakly resolved. We present CoM3D-ACE as a deterministic post-tracking refinement layer. It represents each frozen local tracklet as an EvidenceToken, builds a directed candidate graph with fixed temporal, image-plane, and appearance gates, and retains only reciprocal predecessor-successor proposals. No detector box is added, removed, or interpolated. We evaluate frozen ByteTrack, OC-SORT, and Deep OC-SORT outputs on all 50 sequences of the official MMOT test release. Relative to no refinement, equal-sequence-mean IDF1 rises by 2.28, 2.29, and 2.18 points with common oracle AABBs and by 1.65, 1.62, and 1.00 points with one shared official detector cache. Total identity switches fall by 2,545, 1,678, and 585 and by 1,389, 1,059, and 324, respectively. Paired sequence-bootstrap intervals for these six no-refinement IDF1 differences remain above zero on a locally pre-specified 38-sequence reporting partition. CoM3D-ACE also exceeds a box-preserving AFLink adapter in all six full-test IDF1 comparisons. However, Geometry+ReID greedy and partial-Hungarian assignment attain higher mean IDF1 than the historical CoM3D-ACE ranking in all six conditions; a same-cost control identifies geometry-first ranking as a material limitation. The fixed rule also degrades on held-out M3OT streams. The contribution is therefore an auditable, detector-preserving refinement and diagnostic study, not a claim of universal assignment optimality or cross-domain transfer.

Keywords: aerial multi-object tracking, tracklet refinement, evidence graph, reciprocal association, small aerial objects

*Corresponding authors.

Email addresses: `jerassi@gnu.ac.kr` (Sang-Min Choi), `gunwoo.kim@gnu.ac.kr` (Gun-Woo Kim)

1. Introduction

Aerial video provides wide-area coverage for traffic monitoring, inspection, emergency response, and search-and-rescue, but its targets are frequently small in native image coordinates. In the complete MMOT test release used here, 53.4% of annotated boxes have area below 32^2 pixels and 43.1% have a short side below 16 pixels. At this scale, a short occlusion, missed detection, or weak crop can interrupt a trajectory even when the target reappears only a few frames later. The resulting identity fragmentation affects association metrics and downstream trajectory analysis without necessarily changing localization.

A target is poorly resolved when its projection contains too few sensor samples to preserve class- or identity-discriminative appearance. Enlarging a few-pixel crop cannot recover evidence that the sensor did not record. Across multiple UAV views, the same physical actor may also receive independent view-local identities. Figure 1 illustrates these two motivating failure modes. The present experiments focus on the temporal form of fragmentation within each UAV stream, while the cross-view panel places that problem in the broader cooperative-perception setting.

Modern online trackers attack this problem within the frame-to-frame update. ByteTrack recovers low-score detections during association [1]; OC-SORT improves motion reasoning through observation-centric updates [2]; and Deep OC-SORT supplements that design with appearance cues [3]. A distinct option is to retain these tracker outputs and link compatible tracklets after tracking. Post-processing is attractive because it can be applied to archived results and compared under exactly the same detections and local trajectories. It is also risky: one false link can merge two otherwise valid identities.

This work asks a deliberately narrow question: can a fixed evidence-graph rule improve temporal identity continuity without changing the upstream detector or tracker? We instantiate each local tracklet as an EvidenceToken, form directed candidate relations between temporally ordered fragments, and retain only mutual predecessor-successor choices that pass fixed geometry and appearance gates. A component-level frame-overlap assertion validates the relabeled output. Fragments with no admissible relation remain unchanged. The evaluated MMOT rule uses no camera pose, 3D reconstruction, ground-truth identity, learned link classifier, or box interpolation.

CoM3D-ACE begins after a local tracker has produced tracklets and ends

(a) Poorly resolved tiny-object evidence



(b) Cross-view identity fragmentation

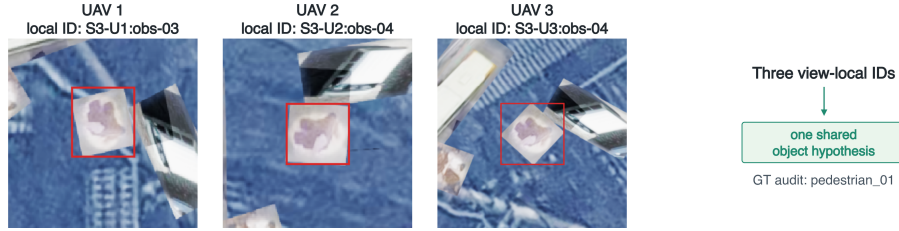


Figure 1: Two failure modes motivating cooperative tiny-object perception. (a) Representative objects in a real VisDrone frame occupy only a few native image pixels, leaving limited appearance detail even after enlargement. Box dimensions report native pixels; enlargement does not create new evidence. (b) Independent UAV views can assign distinct view-local observation identifiers to one physical actor. Actor identity is shown only for this qualitative ground-truth audit. The quantitative study in this paper evaluates the corresponding within-stream temporal-fragmentation problem.

with relabeled temporal trajectories. This separation makes the contribution tracker-agnostic and permits every method to receive exactly the same boxes, classes, confidence values, and local identities before refinement.

The verified strengths of this study are:

- a tracker-agnostic deterministic refinement rule with fixed temporal, image-plane, appearance, and reciprocal constraints, together with explicit output-invariance checks and unchanged output for unsupported fragments;
- protocol-matched evaluation of three frozen tracker families under both common oracle AABs and one shared official detector output on all 50 MMOT test sequences; and
- a locally pre-specified 38-sequence reporting analysis whose six paired

IDF1 intervals against no refinement remain above zero, accompanied by same-input linker, ranking, output-invariance, transfer, and runtime audits.

The contribution is an auditable post-tracking refinement layer that can be applied without detector or tracker retraining, together with a controlled account of where its historical ranking succeeds and fails. Its positive quantitative claim is limited to MMOT improvements over unchanged upstream trajectories and over the validated AFLink adapter; stronger greedy and Hungarian controls and transfer boundaries are reported explicitly.

2. Related Work

2.1. Cooperative aerial perception and small-object evidence

Collaborative perception uses complementary viewpoints before a final scene decision. Where2comm communicates spatially selected features [4], while UCDNet performs feature mapping for cooperative 3D object detection across UAVs [5]. Recent aerial detectors preserve shallow spatial detail or add high-resolution and multi-scale paths for poorly resolved targets [6, 7]. These methods improve upstream sensing or feature fusion. CoM3D-ACE instead starts from object-level outputs and asks whether fragmented evidence can be joined without changing the detector or local tracker. The two roles are complementary: a stronger frontend can supply better observations to the same refinement interface.

2.2. Aerial tracking and evaluation

Aerial MOT combines identity association with small target size, camera motion, viewpoint change, and crowded scenes. MDMT/MIA-Net introduced a multi-drone benchmark and an occlusion-oriented association model [8]; TSMMT models temporal and spatial interactions for multi-drone tracking [9]. MMOT provides multispectral UAV video with oriented annotations and dense, small targets [10], and M3OT adds multi-drone RGB and infrared sequences [11]. The protocol here evaluates individual temporal streams, isolating within-stream tracklet refinement from global cross-camera identity assignment.

ByteTrack associates high- and low-confidence detections [1]. OC-SORT emphasizes observation-centric motion updates under occlusion [2], and Deep OC-SORT introduces adaptive appearance association [3]. We use all three

as frozen upstream trackers rather than modifying their update rules. HOTA decomposes tracking accuracy into detection and association components [12]; IDF1 evaluates identity continuity through global trajectory matching [13]. Both are reported with CLEAR identity switches.

2.3. Post-tracking tracklet linking

Global association can repair trajectories that an online tracker has split. GIAOTracker combines local aerial tracking with a global linking stage [14]. StrongSORT++ includes AFLink, a learned post-linker, and Gaussian-smoothed interpolation as distinct modules [15]. Our comparison reuses assignments from the released AFLink model and checkpoint, excludes interpolation, and applies a separately named output-validity adapter so that every method preserves the common box set. The native post-remap deduplication is audited separately.

Graph-based MOT methods learn structured relations among detections or tracklets. Neural message passing can optimize global associations across a detection graph [16], while city-scale multi-camera systems combine appearance and spatiotemporal constraints [17, 18]. CoM3D-ACE is narrower: it is a deterministic post-processing rule whose refiner and fixed pretrained descriptor receive no target-benchmark training.

2.4. Selective and constrained association

Every post-linker trades recovered continuity against false merges. Selective prediction motivates leaving uncertain cases unresolved rather than forcing a decision [19], but a heuristic distance threshold is not automatically a calibrated risk estimate [20, 21]. We therefore report accepted-link error together with coverage at the fixed operating point and across a descriptive sweep. The component frame-overlap check is a hard output-validity assertion; under the one-shot strict-time reciprocal graph evaluated here, it is structurally redundant rather than an independently active performance module.

Active multi-view methods additionally select informative observations for recognition or detection [22, 23]. That broader decision problem motivates the complete system context shown in Fig. 2; the quantitative claim of this paper remains the fixed temporal-refinement branch defined in Section 3.

3. Method

3.1. System context and evaluated branch

Figure 2 shows the complete CoM3D-ACE object-level interface: synchronized or time-aligned observations enter through an interchangeable frontend, are normalized as EvidenceTokens, and populate an evidence graph whose state can support downstream decisions. The experiments in this paper isolate the path from a frozen local tracker through the EvidenceToken contract and temporal graph to relabeled trajectories. The cross-view hypothesis and active re-observation blocks provide system context; they are not included in the quantitative claim evaluated here.

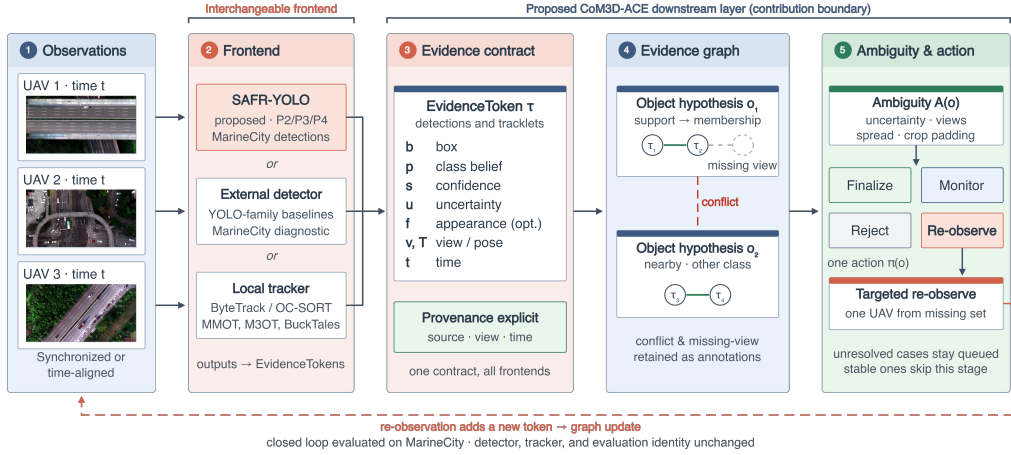


Figure 2: Complete CoM3D-ACE system context and contribution boundary. The architecture accepts detections or local tracklets through one EvidenceToken contract and retains support, conflict, missing-view, and provenance records for downstream decisions. This paper quantitatively evaluates the frozen local-tracker path through temporal evidence-graph refinement; cross-view hypothesis routing and targeted re-observation are shown as broader system context rather than as claims of the present experiment.

3.2. Frozen tracklets as evidence tokens

Let an upstream tracker produce a set of local tracklets $\mathcal{T} = \{T_i\}_{i=1}^K$. Tracklet T_i contains observations $z_{it} = (t, b_{it}, s_{it}, c_i)$, where t is the released frame index, $b_{it} = (x, y, w, h)$ is an axis-aligned box in native image pixels, s_{it} is the upstream confidence, and c_i is the class. In the evaluated adapter, classes are processed independently, so every candidate relation is same-class by construction.

An EvidenceToken for tracklet i stores T_i , its frame support $F_i = \{t : z_{it} \in T_i\}$, first and last endpoint boxes, and an optional appearance descriptor $\mathbf{f}_i \in \mathbb{R}^{128}$. The descriptor is computed from at most three direct tracker-box crops: the first, middle, and last observations. Each crop is resized to 128×256 , transformed by the fixed BaseReID evaluation normalization, and L2-normalized after encoding. We average the available crop features and normalize the result again. In the MMOT protocols, no ground-truth matching is used to select or encode a crop. The historical M3OT oracle-box transfer diagnostic uses a separate crop-admission procedure documented in Section 4.

The full appearance-dependent rule requires a valid descriptor at both edge endpoints. If either descriptor is unavailable because all sampled crops are invalid, that edge is inadmissible; a tracklet with no other admissible relation retains its original identity. We do not impute a zero vector or silently fall back to geometry within the full method.

3.3. Directed candidate graph

A directed edge $i \rightarrow j$ is temporally eligible only if the two frame supports are disjoint, T_i ends before T_j begins, and

$$1 \leq g_{ij} = \min F_j - \max F_i \leq G, \quad G = 30. \quad (1)$$

Let $\mathbf{p}_i^{\text{end}}$ and $\mathbf{p}_j^{\text{start}}$ be the centers of the endpoint boxes. The image-plane and appearance distances are

$$d_{ij}^{xy} = \|\mathbf{p}_i^{\text{end}} - \mathbf{p}_j^{\text{start}}\|_2, \quad d_{ij}^{app} = 1 - \mathbf{f}_i^\top \mathbf{f}_j. \quad (2)$$

The fixed candidate gates are $d_{ij}^{xy} \leq 55$ pixels and $d_{ij}^{app} \leq 0.30$. These values were fixed from M3OT development data before the held-out tests used here. They are not adapted to sequence name, modality, tracker, or MMOT identity labels.

3.4. Reciprocal proposal and component constraint

For every source tracklet, the method chooses the outgoing candidate with the lexicographically smallest tuple

$$q_{ij} = (d_{ij}^{xy}, d_{ij}^{app}, g_{ij}, i, j). \quad (3)$$

It independently chooses the smallest incoming tuple for every destination. An edge becomes a proposal only when it is both choices. This reciprocal test

prevents one source from being attached to a destination that prefers another predecessor, while naturally leaving unsupported tracklets unmatched.

Proposals are processed in ascending $(d_{ij}^{app}, g_{ij}, i, j)$ order with a disjoint-set data structure. Before joining components C_i and C_j , the method checks

$$\left(\bigcup_{u \in C_i} F_u \right) \cap \left(\bigcup_{v \in C_j} F_v \right) = \emptyset. \quad (4)$$

A proposal that violates Eq. (4) is rejected; otherwise the components are joined and inherit the earlier root identity. In the implemented one-shot graph, however, strict forward-time proposals and at most one reciprocal predecessor and successor make every connected component a directed path. Given duplicate-free input tracklets, path members cannot overlap in time. Property tests over all 1,989 MMOT sequence/class instances therefore observe no guard rejection and no guard/no-guard output difference. Equation (4) is retained as a defensive assertion and output validator, not as an independently active performance module.

Finally, only track IDs are remapped. Frame indices, boxes, classes, confidence values, and the number of predictions are invariant. No interpolation fills temporal gaps.

This implementation does not use the candidate score as calibrated uncertainty, and it has no active re-observation action. With E eligible edges, exhaustive candidate construction costs $O(K^2)$, proposal sorting costs $O(E \log E)$, and union-find updates are near-linear in the number of accepted proposals. The explicit candidate graph requires $O(K + E)$ memory.

4. Experimental Protocol

4.1. Datasets, inputs, and frozen split

MMOT. We use all 50 sequences in the official MMOT test release [10]: 5,466 frames and 196,194 valid annotated boxes. Raw images are HWC `uint8` arrays containing eight spectral bands. Following the MMOT paper and the active official tracking loaders, pseudo-RGB is formed from one-based bands 5, 3, and 2, corresponding to zero-based indices $[4, 2, 1]$ in RGB order. The earlier local $[0, 1, 2]$ extraction was not equivalent; all appearance-dependent results in this paper were recomputed with the corrected channels.

Two input conditions are kept separate. The *oracle-AABB* condition converts each official oriented annotation to its axis-aligned envelope, assigns

confidence 1.0, and gives every method the same boxes and classes. Ground-truth identity remains hidden until scoring. The *official detector* condition runs the MMOT authors’ YOLO11L-3ch checkpoint once at 1,280-pixel input, confidence 0.1, and NMS IoU 0.6. Its 193,977 detections over all 5,466 frames, including one frame with no detection, are cached and reused by every tracker and refiner.

Twelve MMOT sequences had appeared in our earlier analysis. A local reporting plan retained these as **legacy12** and assigned every other official test sequence to **confirmation38**; no sequence was selected by performance. The exact membership is listed in Table A.6. The rule and all thresholds remain unchanged across both partitions. Because no externally timestamped preregistration is available, we call this split locally pre-specified rather than independently preregistered.

M3OT transfer.. M3OT [11] contributes four development streams (1-08T, 1-08, 2-08T, and 2-08) and four held-out streams (1-03T, 1-03, 2-03T, and 2-03). Suffix T denotes infrared. The temporal, geometry, appearance, and reciprocal settings were fixed from the development streams before held-out evaluation. Each stream is scored independently with the release’s vehicle filter and common oracle AABBs. To reproduce the historical descriptor cache, tracker observations are admitted to image crops through $\text{IoU} \geq 0.9$ matching to oracle GT before BaseReID encoding. Thus this transfer experiment is an oracle-box diagnostic, not a detector-input result, and GT enters descriptor construction as well as final scoring. We do not use the M3OT held-out result to retune the rule.

Both datasets are used for within-stream temporal identity. Neither protocol evaluates a global identity shared among different UAV cameras.

4.2. Upstream trackers and appearance model

ByteTrack, OC-SORT, and Deep OC-SORT are instantiated through BoxMOT commit **6edfa8ad** [24]. Each official fine-grained MMOT class is processed by an independent tracker instance, every released frame is passed in numeric order, and no area, height, or aspect-ratio filter is added by our adapter. Table 1 records the active settings. They are the historical BoxMOT AABB configurations used by this study, rather than the rotated-tracker settings in MMOT’s native benchmark code.

Tracklet descriptors use a fixed 128-dimensional ResNet-50 checkpoint from the OpenMMLab MOT17 BaseReID configuration. Its SHA-256 is

Table 1: Frozen upstream tracker settings. A dash denotes a parameter not used by that tracker. Deep OC-SORT additionally uses its default adaptive appearance weighting and camera-motion compensation.

Tracker	Det. gate	Match gate	Max age	Min hits	IoU gate	Δt
ByteTrack	0.45	0.80	30	3	0.30	–
OC-SORT	0.30	–	30	3	0.30	3
Deep OC-SORT	0.50	–	30	3	0.30	3

reported in Appendix E. The test transform resizes crops to 128×256 and applies the checkpoint’s ImageNet normalization. The descriptor model and the proposed refiner are not trained or adapted on MMOT or M3OT. The official detector checkpoint is MMOT-trained and is outside that no-target-training statement. Deep OC-SORT receives per-detection embeddings from the same checkpoint; CoM3D-ACE then computes tracklet-level descriptors from the frozen output boxes.

4.3. Comparison methods

All refiners receive exactly the same saved upstream tracklets for a given tracker and input condition.

- **None** leaves every local identity unchanged.
- **Geometry greedy** takes every temporally eligible edge within 55 pixels, orders edges by endpoint distance, and applies the component frame-overlap constraint.
- **Geometry+ReID greedy** applies the same temporal, geometry, and appearance gates as CoM3D-ACE, then processes all gated edges by appearance distance, gap, and deterministic tracklet-ID tie breaks.
- **Partial Hungarian** uses candidates satisfying the same geometry and appearance gates. Its controlled cost is the mean of $d^{xy}/55$, $d^{app}/0.30$, and $g/30$, with a dummy unmatched cost of 1.0. Selected edges also pass the component constraint.
- **AFLink+validity adapter** reuses assignments from the official StrongSORT implementation and checkpoint [15], with released settings $\text{thrT} = (0, 30)$, $\text{thrS} = 75$, and $\text{thrP} = 0.05$. Native AFLink remaps

identities and then keeps only the first row for each duplicate frame–identity pair. For the common-box comparison, a fixed no-GT adapter instead rejects an assigned component merge whenever its frame supports overlap. Gaussian-smoothed interpolation is not applied.

- **CoM3D-ACE** is the reciprocal rule in Section 3.

The ablation additionally evaluates geometry-only reciprocal selection and reciprocal linking without the component assertion. A causal diagnostic holds the gated candidate graph and scalar cost fixed while changing only greedy, reciprocal, or partial-Hungarian assignment; this diagnostic is not used to replace the historical method after observing test performance. GIAOTracker is discussed as relevant global-linking work [14], but its authors’ public repository states that code release is not planned and does not provide the learned global-link model and weights needed for a faithful run. We therefore do not label a local approximation as official GIAOTracker.

4.4. Metrics and aggregation

TrackEval commit `12c8791b` computes HOTA, its association component AssA, IDF1, CLEAR MOTA, and identity switches (IDSW). HOTA averages the standard localization-threshold curve; AssA measures association quality among matched detections. IDF1 is the F1 score after global one-to-one trajectory matching, and IDSW counts changes in the matched predicted identity over time. Higher is better for HOTA, AssA, IDF1, and MOTA; lower is better for IDSW.

Headline HOTA, AssA, and IDF1 values are equal-sequence means, so long sequences do not dominate. IDSW, FP, and FN are summed over exactly the same sequence set. We never recompute F1 from averaged precision and recall. Paired differences against no refinement and the principal same-input linkers use 10,000 released-sequence bootstrap resamples with seed 20260914. The official release metadata available to us does not map sequence identifiers to source flights, so we do not infer groups from filename prefixes or report an unsupported flight-level block bootstrap. Repeated deterministic executions are reproducibility checks, not independent statistical samples.

For the link audit, an endpoint tracklet receives a post-hoc actor label only when its matched observations have a unique majority identity. Link coverage uses the fixed set of source tracklets with at least one temporally eligible, non-overlapping successor within 30 frames. Link error is the false-link fraction

among accepted edges whose two endpoint identities are auditable. Unknown edges remain in coverage but not in the error denominator. The sweep is descriptive and is not used to select a test threshold.

4.5. Integrity and runtime protocol

Before evaluation, exact GT-as-prediction and globally permuted-ID tests had to return HOTA, IDF1, and MOTA of 100 with IDSW 0. No-refinement had to reproduce the cached tracker output exactly. For every headline refiner, we check preservation of the (`frame`, `box`, `confidence`, `class`) multiset and reject aggregate reporting when one identity appears twice in a frame. Ground-truth actor identity is read only after MMOT predictions are frozen, for metrics and post-hoc edge labels.

Runtime is measured over the full MMOT test set on an NVIDIA RTX 5090 with PyTorch 2.9.1 and CUDA 12.8. BaseReID uses batch 128. Graph timing excludes disk I/O, upstream tracking, descriptor extraction, and metric evaluation; total post-tracking overhead adds descriptor and graph time and divides by 5,466 frames. These are offline replay measurements rather than flight-system latencies.

5. Results

5.1. Full-test same-input comparison

Table 2 places all executable principal linkers in the main same-input comparison. For a given input and upstream tracker, every row starts from identical boxes, classes, confidences, and local tracklets. Relative to no refinement, CoM3D-ACE raises equal-sequence-mean IDF1 by 2.28, 2.29, and 2.18 percentage points for oracle ByteTrack, OC-SORT, and Deep OC-SORT, respectively, while total IDSW decreases by 2,545, 1,678, and 585. With the shared official detector cache, the corresponding IDF1 increases are 1.65, 1.62, and 1.00 points, with 1,389, 1,059, and 324 fewer switches. These deltas are calculated from the full-precision sequence results before display rounding.

The broader comparison changes the interpretation. CoM3D-ACE has higher IDF1 than the box-preserving AFLink adapter in all six conditions; the paired all-50 intervals for these differences are also above zero. Geometry+ReID greedy and partial Hungarian, however, have higher mean IDF1 than the historical CoM3D-ACE ranking in all six conditions. Their paired IDF1 intervals against CoM3D-ACE are entirely above zero in their favor in five conditions; only the official-detector Deep OC-SORT interval includes

zero (Table B.9). CoM3D-ACE is therefore not claimed as the most accurate assignment rule.

Table 2: Principal same-input MMOT comparison over all 50 official test sequences. IDF1 is the equal-sequence mean and IDSW is the total. Every refiner receives the same frozen local tracklets; bold numbers mark the best value in each tracker/input column. AFLink assignments use the separately named no-GT validity adapter (VA) described in Section 4; Geo. abbreviates geometry and Hung. abbreviates Hungarian.

Input	Refiner	ByteTrack		OC-SORT		Deep OC-SORT	
		IDF1	IDSW	IDF1	IDSW	IDF1	IDSW
Oracle AABB	None	47.44	20,056	42.96	6,119	81.49	1,962
	Geometry greedy	51.02	15,484	45.41	4,122	83.69	1,294
	Geo.+ReID greedy	51.80	15,064	46.25	3,481	83.99	1,149
	Partial Hung.	50.78	16,462	46.19	3,547	84.00	1,174
	AFLink+VA	47.70	20,025	43.33	6,082	81.93	1,919
	CoM3D-ACE	49.72	17,511	45.25	4,441	83.67	1,377
Official detector	None	38.32	10,324	36.46	4,263	63.09	1,370
	Geometry greedy	40.76	7,908	38.14	2,959	64.01	1,008
	Geo.+ReID greedy	41.10	7,711	38.76	2,585	64.15	942
	Partial Hung.	40.56	8,405	38.74	2,668	64.17	957
	AFLink+VA	38.53	10,297	36.69	4,240	63.33	1,342
	CoM3D-ACE	39.97	8,935	38.09	3,204	64.08	1,046

The source-stage AFLink audit explains the adapter rather than hiding invalid runs. Frozen StrongSORT source calls native `deduplicate` after identity remapping. Across the six full-test conditions, selected merges create 13 sequence/tracker/class overlap contexts, and native deduplication removes 15 boxes. Decimal save formatting changes numeric serialization separately. The common validity adapter rejects only those overlapping component merges, without GT, and preserves every input box; its valid full-set results appear in Table 2.

The positive findings that remain directly supported are:

- **Upstream preservation:** CoM3D-ACE changes only identity labels; every frame, box, class, confidence, and prediction count is unchanged.
- **No-refinement and AFLink comparisons:** mean identity metrics improve across three tracker families and both input conditions.
- **Protocol breadth:** the direction against no refinement holds with common oracle boxes and one shared official detector output.

- **Training scope:** the refiner is fixed and BaseReID is not trained or adapted on MMOT; this statement does not include the official MMOT-trained detector checkpoint.

5.2. Locally pre-specified reporting partition

The 38 sequences outside the earlier 12-sequence analysis form a locally pre-specified reporting partition. Table 3 reports CoM3D-ACE minus no-refinement changes. All six paired IDF1 intervals remain above zero. With oracle AABBs, 38 and 37 of 38 ByteTrack and OC-SORT sequences improve; with detector inputs, 33, 36, and 27 sequences improve for ByteTrack, OC-SORT, and Deep OC-SORT. These results show that the mean no-refinement gain is not confined to legacy12. Because an externally timestamped preregistration is unavailable, we do not describe this partition as an independent preregistered confirmation set.

Table 3: CoM3D-ACE minus no-refinement paired changes on the locally pre-specified confirmation38 partition. Confidence intervals are 95% released-sequence bootstrap intervals with 10,000 resamples and seed 20260914. The final column gives improved/tied/worsened IDF1 sequence counts.

Input	Tracker	Δ HOTA	Δ AssA	Δ IDF1 (95% CI)	Δ IDSW	I/T/W
Oracle	ByteTrack	+1.05	+1.71	+2.38 [1.85, 2.99]	-2,147	38/0/0
	OC-SORT	+0.91	+2.07	+2.15 [1.54, 2.84]	-1,185	37/0/1
	Deep OC-SORT	+1.28	+2.49	+2.29 [1.37, 3.57]	-466	31/4/3
Official det.	ByteTrack	+0.79	+1.48	+1.71 [1.20, 2.26]	-1,142	33/1/4
	OC-SORT	+0.58	+1.46	+1.51 [1.07, 2.04]	-759	36/0/2
	Deep OC-SORT	+0.44	+0.92	+1.10 [0.48, 2.00]	-268	27/5/6

5.3. Ranking, solver, and transfer diagnostics

The historical CoM3D-ACE row combines reciprocal selection with the geometry-first lexicographic key in Eq. (3). To separate ranking from solver effects, we reuse each frozen candidate graph and the partial-Hungarian scalar cost with greedy, reciprocal, and Hungarian assignment. At the unchanged 0.30 appearance gate, common-cost reciprocal selection raises IDF1 over historical CoM3D-ACE in all six conditions. It also accepts a larger fraction of the fixed eligible-source denominator while reducing known-link error in all six (Tables B.10 and C.12). Thus geometry-first ranking is a material

limitation; the result does not justify renaming the post-hoc common-cost row as the submitted method.

The reciprocal path invariant was also checked directly. Across 1,989 MMOT sequence/class/tracker/input instances, maximum proposal indegree and outdegree are one, with no temporal violation, cycle, duplicate-frame input, component rejection, or guard/no-guard output difference. This confirms that the component check is defensive under the implemented one-shot graph rather than a measured source of accuracy gain.

M3OT provides an adverse transfer diagnostic. The historical rows reproduce exactly, including accepted-edge sets. On held-out streams, CoM3D-ACE accepts 0 correct and 7 false links for ByteTrack and 0 correct and 3 false links for OC-SORT. Among the eight auditable same-actor opportunities, five fail the 55-pixel gate, two lose the reciprocal incoming rank, and one loses the outgoing rank. Common-cost reciprocal recovers one correct link per tracker and reduces false links to 6 and 2, but IDF1 remains below no refinement (Table D.14). The failure therefore involves both ranking and non-transferring cues, not only the choice of reciprocal versus Hungarian assignment.

5.4. Small-object context and execution cost

Table 4 verifies that the evaluation contains many small aerial targets. Across all 50 sequences, 53.4% of boxes have area below 32^2 pixels, 45.9% lie from 32^2 to 96^2 , and the median short side is 21 pixels. This distribution motivates the setting but is not a scale-stratified claim that CoM3D-ACE improves tiny objects more than larger objects.

Table 4: Native MMOT annotation-box distribution. Area bins are fixed COCO boundaries; percentages use all valid boxes in each partition.

Partition	Boxes	$A < 32^2$	$32^2 \leq A < 96^2$	Side < 16	Median side
All 50	196,194	53.4%	45.9%	43.1%	21 px
Legacy 12	30,666	47.9%	51.7%	38.5%	23 px
Confirm. 38	165,528	54.4%	44.9%	44.0%	19 px

Post-tracking runtime is dominated by appearance extraction (Table 5). BaseReID plus graph processing costs 46.9 ms/frame for ByteTrack and approximately 24.5 ms/frame for OC-SORT and Deep OC-SORT. Graph construction and decisions alone total 7.99, 2.09, and 1.21 seconds, respectively, over the 5,466-frame release. A separate same-machine CPU replay shows

lower reciprocal solver time and peak memory than Hungarian in most conditions, but the absolute solver difference is small relative to candidate construction and descriptor extraction (Table C.13). It is not an end-to-end runtime advantage.

Table 5: Full-test post-tracking runtime. ReID uses an RTX 5090 and batch 128; graph time is in-memory CPU candidate construction and decision time. The final column combines ReID and graph time over 5,466 frames.

Tracker	Crops	Desc.	ReID (s)	Cand.	Graph (s)	Total (ms/fr.)
ByteTrack	29,281	13,836	248.3	864,743	7.99	46.9
OC-SORT	15,819	8,150	131.6	216,508	2.09	24.5
Deep OC-SORT	15,490	5,663	132.9	76,253	1.21	24.5

6. Discussion

The clearest positive result is a protocol-matched improvement over unchanged upstream trajectories. CoM3D-ACE raises the equal-sequence means of HOTA, AssA, and IDF1 and lowers total IDSW for three tracker families under both oracle and official-detector inputs. Because the method preserves the complete detection multiset, these changes are attributable to identity relabeling rather than detector retraining, interpolation, or sequence filtering. The locally pre-specified confirmation³⁸ analysis shows that this mean gain is not confined to the 12 sequences used in the earlier analysis.

That finding is different from superiority over other linkers. The restored main comparison and paired intervals show that Geometry+ReID greedy and partial Hungarian produce higher mean IDF1 than the historical CoM3D-ACE ranking in all six conditions. Moreover, replacing only geometry-first ranking with the common scalar cost improves reciprocal assignment itself. The evidence therefore supports reciprocal selection as an auditable deterministic mechanism, but not the historical rank key as an accuracy- or risk-optimal design.

The EvidenceToken abstraction remains operationally useful because it decouples refinement from a particular tracker implementation. ByteTrack, OC-SORT, and Deep OC-SORT expose different motion and appearance behavior, yet the same candidate graph can process their archived tracklets. Unsupported fragments remain unchanged, every merge is recoverable from

an edge trace, and the native AFLink failure illustrates why preserving an explicit output contract matters.

The transfer and runtime audits further delimit this role. M3OT shows that a fixed pixel gate and MOT17 appearance descriptor do not transfer reliably, even when a different solver recovers some correct candidates. Appearance extraction and candidate construction also dominate assignment time, so a modest reciprocal-versus-Hungarian solver saving is not a practical end-to-end advantage. These findings motivate future development-set redesign while keeping the present test results unchanged.

7. Limitations

The verified claim is deliberately bounded to deterministic temporal refinement of frozen MMOT trajectories. Figure 2 shows the broader CoM3D-ACE interface, but the present protocol does not evaluate a persistent identity shared across UAV cameras, 3D reconstruction, coordinated view planning, or active re-observation. These functions require their own identity and action-level ground truth before they can support quantitative claims.

The method is not a universally optimal assignment rule. Geometry+ReID greedy and partial Hungarian have higher full-test mean IDF1 than historical CoM3D-ACE in all six same-input conditions. Common-cost reciprocal assignment also improves both IDF1 and the observed error-coverage pair over the historical geometry-first ranking in all six conditions. Accordingly, the current contribution is an auditable fixed refiner and controlled diagnosis, not a state-of-the-art linker claim. Any revised ranking must be designed on development data and evaluated on new confirmation data rather than selected from these test results.

The component-overlap check is a defensive assertion rather than an independent performance module. Strict forward-time reciprocal proposals form disjoint paths when input tracklets contain no duplicate frame, so the check cannot reject a proposal under the evaluated one-shot rules. Accepted-link error is also defined only where both endpoint identities are auditable; unknown links remain in coverage. These descriptive rates are not calibrated risk estimates.

Cross-dataset transfer remains unresolved. The unchanged MMOT-beneficial rule improves four M3OT development streams but degrades four held-out streams. Five of eight held-out same-actor opportunities fail the fixed pixel gate, and common-cost reciprocal selection recovers only one correct

link per tracker while remaining below no refinement. The M3OT descriptor cache also uses oracle-GT IoU crop admission, which limits it to a diagnostic rather than a detector-input transfer claim.

Native AFLink deduplication resolves overlapping remapped identities by deleting boxes and therefore does not satisfy the common-box contract. We report a separately named no-GT validity adapter instead of calling it unmodified official output. A faithful GIAOTracker run remains blocked by unavailable released implementation and compatible weights. Finally, the MMOT release metadata available to us does not expose source-flight groupings, the local confirmation split lacks external preregistration evidence, and runtime is an offline replay on one workstation rather than an onboard latency or energy claim.

8. Conclusion

CoM3D-ACE provides a deterministic evidence-graph refiner for frozen aerial tracklets. Relative to no refinement, it improves equal-sequence-mean HOTA, AssA, and IDF1 and reduces total identity switches over 50 MMOT sequences for ByteTrack, OC-SORT, and Deep OC-SORT under both oracle and official-detector inputs. The no-refinement IDF1 difference remains positive on the locally pre-specified 38-sequence partition, and every input box, class, confidence, and prediction count is preserved.

The completed controls also establish the boundary of that result. Stronger Geometry+ReID greedy and partial-Hungarian rows exceed the historical CoM3D-ACE mean IDF1, a common-cost test identifies geometry-first ranking as a limitation, and held-out M3OT transfer is negative. CoM3D-ACE should therefore be read as a reproducible, auditable MMOT refinement and diagnostic framework, not as a universally superior association rule. Cross-camera identity, active observation, and redesigned cross-domain ranking remain future work requiring new, independently evaluated protocols.

CRediT authorship contribution statement

Jong-Chan Park: Conceptualization, Methodology, Investigation, Writing – original draft. So-Hee Lim: Visualization, Writing – review & editing. Sujin Park: Visualization, Writing – review & editing. Sang-Min Choi: Formal analysis, Writing – review & editing. Gun-Woo Kim: Formal analysis, Writing – review & editing.

Data and code availability

The MMOT [10] and M3OT [11] datasets are publicly available from their official sources: the MMOT repository, the MMOT dataset page, and the M3OT repository. The implementation, frozen split manifest, configurations, and reproduction instructions are planned for public release through a GitHub repository. No repository URL is claimed before that release is available.

Appendix A. MMOT reporting partitions

Table A.6 records the locally pre-specified reporting split. The confirmation38 partition contains every official test sequence not in the previously analyzed legacy12 set.

Table A.6: Locally pre-specified MMOT reporting partition. Frame counts and source hashes are stored in the accompanying split manifest. No sequence was assigned by performance; no external preregistration is claimed.

Partition	Sequence identifiers
Legacy12	data28-1, data28-2, data28-3, data28-4, data28-5, data28-6, data30-2, data30-3, data30-4, data30-5, data30-9, data30-10
Confirm. 38	data23-1, data24-1, data27-1, data31-1, data33-1, data34-1, data34-2, data34-3, data36-4, data36-5, data36-10, data36-11, data36-12, data36-13, data37-1, data37-2, data37-10, data37-11, data37-12, data39-1, data39-2, data39-3, data39-6, data40-3, data40-4, data40-5, data41-1, data42-1, data42-2, data42-3, data46-11, data46-12, data47-1, data47-2, data47-3, data47-4, data48-1, data49-2

Appendix B. Extended comparison controls

Tables B.7 and B.8 provide HOTA and AssA in addition to the compact principal comparison in Table 2. All rows receive the same saved tracklets within each upstream/input condition.

Table B.9 reports paired all-50 differences rather than relying only on aggregate means. Negative values favor the comparator.

Table B.10 isolates ranking from assignment. Historical R is the submitted geometry-first reciprocal row. Cost-G, Cost-R, and Cost-H use the same gated

Table B.7: MMOT full-test temporal refinement with common oracle AABBs. HOTA, AssA, and IDF1 are equal-sequence means over 50 sequences; IDSW is the total. Bold values mark the best valid refiner for each upstream tracker. AFLink+VA uses the fixed no-GT output-validity adapter described in Section 4; Geo. and Hung. abbreviate geometry and Hungarian.

Upstream	Refiner	HOTA	AssA	IDF1	IDSW
ByteTrack	None	44.20	44.69	47.44	20,056
	Geometry greedy	45.60	46.97	51.02	15,484
	Geo.+ReID greedy	45.89	47.47	51.80	15,064
	Partial Hung.	45.55	46.90	50.78	16,462
	AFLink+VA	44.30	44.88	47.70	20,025
	CoM3D-ACE	45.19	46.34	49.72	17,511
OC-SORT	None	44.92	51.43	42.96	6,119
	Geometry greedy	45.90	53.68	45.41	4,122
	Geo.+ReID greedy	46.31	54.55	46.25	3,481
	Partial Hung.	46.26	54.45	46.19	3,547
	AFLink+VA	45.14	51.92	43.33	6,082
	CoM3D-ACE	45.92	53.64	45.25	4,441
Deep OC-SORT	None	81.32	82.06	81.49	1,962
	Geometry greedy	82.58	84.52	83.69	1,294
	Geo.+ReID greedy	82.82	84.98	83.99	1,149
	Partial Hung.	82.79	84.93	84.00	1,174
	AFLink+VA	81.54	82.48	81.93	1,919
	CoM3D-ACE	82.60	84.54	83.67	1,377

candidate graph and scalar cost with greedy, reciprocal, and partial-Hungarian assignment, respectively.

Table B.8: MMOT full-test temporal refinement from one shared official YOLO11L-3ch detection cache. Aggregation and invariant handling match Table B.7.

Upstream	Refiner	HOTA	AssA	IDF1	IDSW
ByteTrack	None	34.65	42.29	38.32	10,324
	Geometry greedy	35.64	44.14	40.76	7,908
	Geo.+ReID greedy	35.79	44.49	41.10	7,711
	Partial Hung.	35.58	44.11	40.56	8,405
	AFLink+VA	34.73	42.45	38.53	10,297
	CoM3D-ACE	35.40	43.75	39.97	8,935
OC-SORT	None	35.89	47.62	36.46	4,263
	Geometry greedy	36.51	49.40	38.14	2,959
	Geo.+ReID greedy	36.80	50.12	38.76	2,585
	Partial Hung.	36.77	49.97	38.74	2,668
	AFLink+VA	36.02	47.95	36.69	4,240
	CoM3D-ACE	36.53	49.24	38.09	3,204
Deep OC-SORT	None	53.92	67.00	63.09	1,370
	Geometry greedy	54.25	67.71	64.01	1,008
	Geo.+ReID greedy	54.34	67.92	64.15	942
	Partial Hung.	54.36	67.95	64.17	957
	AFLink+VA	54.00	67.16	63.33	1,342
	CoM3D-ACE	54.31	67.84	64.08	1,046

Table B.9: CoM3D-ACE minus principal same-input linker IDF1, with 95% paired released-sequence bootstrap intervals (10,000 resamples; seed 20260914). The official-detector Deep OC-SORT intervals include zero; the other ten intervals are below zero.

Input	Tracker	vs. Geo.+ReID greedy	vs. partial Hung.
Oracle	ByteTrack	−2.08 [−2.69, −1.53]	−1.07 [−1.37, −0.79]
	OC-SORT	−1.00 [−1.38, −0.66]	−0.93 [−1.36, −0.57]
	Deep OC-SORT	−0.32 [−0.50, −0.15]	−0.33 [−0.51, −0.17]
Detector	ByteTrack	−1.14 [−1.85, −0.60]	−0.60 [−0.82, −0.40]
	OC-SORT	−0.68 [−1.01, −0.41]	−0.66 [−0.99, −0.40]
	Deep OC-SORT	−0.06 [−0.14, +0.01]	−0.09 [−0.19, +0.01]

Table B.10: Cache-controlled ranking/solver diagnostic on all 50 MMOT sequences. Each cell is equal-sequence-mean IDF1/total IDSW. The fixed 0.30 appearance gate is retained; no test threshold is selected.

Input	Tracker	Hist. R	Cost-G	Cost-R	Cost-H
Oracle	ByteTrack	49.72/17,511	51.54/14,929	50.30/16,710	50.78/16,462
	OC-SORT	45.25/4,441	46.14/3,501	45.88/3,828	46.19/3,547
	Deep OC-SORT	83.67/1,377	83.96/1,185	84.02/1,242	84.00/1,174
Detector	ByteTrack	39.97/8,935	40.97/7,604	40.42/8,491	40.56/8,405
	OC-SORT	38.09/3,204	38.66/2,635	38.55/2,825	38.74/2,668
	Deep OC-SORT	64.08/1,046	64.17/945	64.23/979	64.17/957

Appendix C. Component and accepted-link diagnostics

Table C.11 separates the fixed graph inputs. Geometry-only reciprocal selection accounts for most of the no-refinement gain, and the appearance gate adds a smaller increment. The no-guard and full rows are identical for the structural path reason established in Section 3, not because an independently active module happens to have no effect.

Table C.11: Temporal component ablation on all 50 MMOT sequences with oracle AABBs. IDF1 is the equal-sequence mean and IDSW is the total. All rows reuse the same upstream tracklets and candidate universe.

Variant	ByteTrack		OC-SORT		Deep OC-SORT	
	IDF1	IDSW	IDF1	IDSW	IDF1	IDSW
None	47.44	20,056	42.96	6,119	81.49	1,962
Geometry reciprocal	49.62	17,581	45.08	4,551	83.63	1,392
+ReID reciprocal, no guard	49.72	17,511	45.25	4,441	83.67	1,377
CoM3D-ACE	49.72	17,511	45.25	4,441	83.67	1,377

Table C.12 reports the fixed operating point for the historical and common-cost reciprocal rankings. Coverage uses the fixed eligible-source denominator; error uses only accepted links whose endpoint identities are auditable. Unknown endpoint labels remain in coverage but not in the error denominator. The rates are descriptive and do not define calibrated risk.

Table C.12: Full-test link audit at the unchanged 0.30 appearance gate. Coverage uses the same eligible-source denominator. Link error is false/(correct+false) for auditable endpoints; unknown accepted links remain in coverage but not the error denominator. C/F denotes correct/false accepted links. Cost-R is a post-hoc causal diagnostic, not a selected replacement method.

Input	Tracker	Rule	Coverage	Error	Known C/F	Unknown
Oracle	ByteTrack	Hist. R	36.6%	44.6%	317/255	3,909
		Cost-R	39.8%	43.2%	369/281	4,226
	OC-SORT	Hist. R	37.9%	35.1%	1,661/899	0
		Cost-R	42.1%	21.2%	2,239/602	0
	Deep OC-SORT	Hist. R	26.1%	39.2%	575/371	0
		Cost-R	27.4%	29.0%	704/287	0
Detector	ByteTrack	Hist. R	34.4%	29.7%	97/41	2,796
		Cost-R	37.2%	24.8%	106/35	3,033
	OC-SORT	Hist. R	35.4%	38.9%	173/110	1,766
		Cost-R	39.0%	31.6%	206/95	1,954
	Deep OC-SORT	Hist. R	21.7%	32.3%	42/20	675
		Cost-R	22.1%	28.8%	47/19	686

Table C.13: Cache-only solver replay on an Intel Xeon Silver 4510 CPU. Times sum instance means over the full release; peak memory is the maximum Python allocation traced for one solver call. Historical R and Cost-H reuse the same candidate graph. Cand. is candidate-graph construction. Descriptor extraction and AFLink model inference are excluded.

Input	Tracker	Cand. (ms)	Hist. R (ms/MB)	Cost-H (ms/MB)
Oracle	ByteTrack	8,263.7	191.5/3.9	325.8/17.9
	OC-SORT	1,965.0	82.9/3.6	102.1/3.6
	Deep OC-SORT	1,086.7	118.1/6.6	127.9/6.7
Detector	ByteTrack	3,153.8	133.7/4.2	184.4/7.1
	OC-SORT	1,408.3	77.8/3.8	92.8/3.8
	Deep OC-SORT	817.1	95.7/4.3	101.7/4.4

Appendix D. Cross-dataset transfer diagnostic

The same temporal, image-plane, appearance, and reciprocal settings were fixed on four M3OT development streams and then applied unchanged to four held-out streams. The historical outputs and edge sets reproduce exactly. Of eight held-out same-actor opportunities before the geometry/appearance gates, five fail the geometry gate, two lose incoming rank, and one loses outgoing rank. Table D.14 compares the unchanged inputs under alternative

solvers without retuning.

Table D.14: Fixed-policy M3OT development and held-out diagnostic. The upper table reports equal-sequence-mean IDF1/total IDSW; the lower reports post-hoc accepted correct/false links. Historical R is geometry-first reciprocal; Cost-G, Cost-R, and Cost-H share the same scalar cost. The descriptor cache uses oracle-GT IoU crop admission.

Split	Tracker	None	Historical R	Cost-G	Cost-R	Cost-H
Development	ByteTrack	96.89/20	98.49/10	98.47/11	98.47/11	98.55/9
	OC-SORT	96.10/21	98.26/9	98.24/10	98.24/10	98.33/6
Held-out	ByteTrack	96.77/6	92.77/6	92.89/5	92.89/5	91.86/6
	OC-SORT	95.39/5	94.84/5	94.92/4	94.92/4	93.96/5

Split	Tracker	Historical R	Cost-G	Cost-R	Cost-H
Development	ByteTrack	10/3	9/5	9/4	11/4
	OC-SORT	12/5	11/7	11/6	15/4
Held-out	ByteTrack	0/7	1/6	1/6	0/8
	OC-SORT	0/3	1/2	1/2	0/4

Appendix E. Reproducibility record

The release manifest records full revisions and checksums. Their compact prefixes are:

- MMOT test revision: 57dfe1521ee2;
- BaseReID SHA-256: 4bf6b63d4923;
- MMOT YOLO11L-3ch SHA-256: 6d9f66ed17eb;
- AFLink SHA-256: b35cbecddd3ac;
- TrackEval commit: 12c8791b303e;
- BoxMOT commit: 6edfa8ad7dc2.

Evaluator sanity tests return HOTA/IDF1/MOTA 100 and IDSW 0 for both exact GT predictions and a global one-to-one permutation of prediction identities. Legacy12 rows in the standalone and full-50 executions match at absolute tolerance zero. The source artifact package contains commands, source and model hashes, successful manifests, failed-run records, per-sequence CSVs, and a SHA-256 index.

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